Elligator

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June 26, 2026

Chapter 1

Elligator 1

Theorem 1 (u defined)**.** *The following element of* $\mathbb{F}_q$ *is defined for each* $t \in \mathbb{F}_q \setminus \{-1, 1\}$*:*

$$u = (1 - t)/(1 + t),$$

Proof. □

Theorem 2 (Y defined)**.** *The following element of* $\mathbb{F}_q$ *is defined for each* $t \in \mathbb{F}_q \setminus \{-1, 1\}$*:*

$$Y = (\chi(v)v)^{(q+1)/4}\chi(v)\chi(u^2 + 1/c^2),$$

Proof. □

Theorem 3 (x defined)**.** *The following element of* $\mathbb{F}_q$ *is defined for each* $t \in \mathbb{F}_q \setminus \{-1, 1\}$*:*

$$x = (c - 1)sX(1 + X)/Y,$$

Proof. □

Theorem 4 (y defined)**.** *The following element of* $\mathbb{F}_q$ *is defined for each* $t \in \mathbb{F}_q \setminus \{-1, 1\}$*:*

$$y = (rX - (1 + X)^2)/(rX + (1 + X)^2).$$

Proof. □

Theorem 5 (Fulfill Helper Curve Equation)**.** *Let* r*,* X *and* Y *as defined above.*
Then $Y^2 = X^5 + (r^2 - 2)X^3 + X$ *holds.*

Proof. □

Theorem 6 (Non zero helper variables)**.** *Let* u*,* v*,* X*,* Y*,* x*,* y *as defined above.*
Then $uvXYx(y + 1) \neq 0$*.*

Proof. □

Theorem 7 (Fulfill Curve Equation)**.** *Let* x *and* y *as defined above.*
Then $x^2 + y^2 = 1 + dx^2y^2$ *holds.*

Proof. □

Definition 8 (The Map)**.** Definition 2. In the situation of Theorem 1, the decoding function for the complete Edwards curve $E : x^2 + y^2 = 1 + dx^2y^2$ is the function $\varphi : \mathbb{F}_q \to E(\mathbb{F}_q)$ defined as follows:

$$\varphi(\pm 1) = (0, 1);$$

if $t \notin \{\pm 1\}$ then

$$\varphi(t) = (x, y).$$

Original: Chapter "3.2 The map": Definition 2

Theorem 9.

Proof. □

Theorem 10.

Proof. □

Theorem 11.

Proof. □

Definition 12 (The Encoding Function)**.** In the situation of Definition 2, assume that q is prime, and define $b = \lfloor \log_2 q \rfloor$. Define $\sigma : \{0, 1\}^b \to \mathbb{F}_q$ by

$$\sigma(\tau_0, \tau_1, \dots, \tau_{b-1}) = \sum_i \tau_i 2^i.$$

Define

$$S = \sigma^{-1}(\{0, 1, 2, \dots, (q - 1)/2\}).$$

Define $\iota : S \to E(\mathbb{F}_q)$ as follows:

$$\iota(\tau) = \varphi(\sigma(\tau)).$$

Original: Chapter "3.4 Encoding as strings": Theorem 4

Theorem 13 (Cardinality of S)**.** *In the situation of Definition 2, assume that q is prime, and define $b = \lfloor \log_2 q \rfloor$. Define $\sigma : \{0, 1\}^b \to \mathbb{F}_q$ by*

$$\sigma(\tau_0, \tau_1, \dots, \tau_{b-1}) = \sum_i \tau_i 2^i.$$

Define

$$S = \sigma^{-1}(\{0, 1, 2, \dots, (q - 1)/2\}).$$

Then $\#S = (q + 1)/2$;
Original: Chapter "3.4 Encoding as strings": 1. statement of Theorem 4

Proof. □

Theorem 14 (Cardinality of S)**.** *In the situation of Definition 2, assume that q is prime, and define $b = \lfloor \log_2 q \rfloor$. Define $\sigma : \{0, 1\}^b \to \mathbb{F}_q$ by*

$$\sigma(\tau_0, \tau_1, \dots, \tau_{b-1}) = \sum_i \tau_i 2^i.$$

Define

$$S = \sigma^{-1}(\{0, 1, 2, \dots, (q - 1)/2\}).$$

Define $\iota : S \to E(\mathbb{F}_q)$ as follows:

$$\iota(\tau) = \varphi(\sigma(\tau)).$$

Then $\#S = (q+1)/2$;
Original: Chapter "3.4 Encoding as strings": 1. statement of Theorem 4

Proof. □

Theorem 15.

Proof. □